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Doctorado en Ingeniería Física





Geomagnetic Activity and Reported Aurora Sightings

Analysis Using Python and Power BI

Final Integrative Project (PIA)

Scientific Computing (D06)

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Abstract

This project investigates the relationship between geomagnetic activity and reported aurora observations during the year of 2024. The planetary Kp index was obtained from the GFZ scientific database. Historical aurora observation records were collected from Aurorasaurus. And NASA DONKI geomagnetic storm records were also included as supporting data.

Python was used to retrieve, process, and analyze the datasets. Power BI was used to create graphs and compare the results.

The analysis included 2,928 Kp measurements and 11,465 Aurorasaurus observation records. Pearson and Spearman correlation coefficients of 0.4498 and 0.6181 were obtained, respectively.

The results suggest a positive association between geomagnetic disturbances and the number of reported aurora observations. A separate analysis of October 2026 was also included to demonstrate the application of the same computational workflow.

1 Introduction

Earth has a magnetic field that is mainly generated by the movement of electrically conducting liquid iron in its outer core. This magnetic field extends into space and forms the magnetosphere, which interacts with charged particles and magnetic fields coming from the Sun.

The Sun continuously releases a stream of plasma known as the solar wind. Changes in solar wind conditions can disturb Earth's magnetosphere. During stronger disturbances, variations can be observed in Earth's magnetic field.

One way to describe these disturbances is through the planetary Kp index. Kp is a dimensionless geomagnetic activity index ranging from 0 to 9. It is calculated using measurements from ground-based magnetic observatories and is reported every three hours.

A low Kp value represents relatively quiet geomagnetic conditions, while higher values indicate stronger disturbances. Values of $Kp \geq 5$ represent storm-level geomagnetic activity.

The Kp index does not measure the absolute strength of Earth's magnetic field. Instead, it describes the level of geomagnetic disturbance.

Auroras are another phenomenon associated with interactions between the solar wind and Earth's magnetosphere.

They form when energetic charged particles, mainly electrons, enter the upper atmosphere and interact with gases such as oxygen and nitrogen. These interactions produce the visible light commonly known as the Northern and Southern Lights.

Stronger geomagnetic disturbances can increase the possibility of observing auroras over a wider geographical area.

However, auroral visibility also depends on weather, darkness, geographic location, and other conditions.

This project examines whether higher geomagnetic activity is associated with an increase in reported aurora observations.

1.1 Project Objective

The main objective is to analyze geomagnetic activity during 2024 and investigate its relationship with reported aurora observations using Python and Power BI.

The study focuses on the following Scientific Question:

> *Are days with higher geomagnetic Kp associated with more reported aurora observations during 2024?*

2 Data and Methodology

The project uses three scientific data sources: GFZ, Aurorasaurus, and NASA DONKI.

2.1 Data Sources

GFZ Kp Data

The geomagnetic data were obtained from the GFZ Helmholtz Centre for Geosciences using its public scientific API.

The dataset covers January 1 to December 31, 2024, with one Kp measurement every three hours.

Since 2024 was a leap year, the dataset contains 366 days and 2,928 observations.

Aurorasaurus Data

The aurora observation dataset was obtained from the historical cleaned Aurorasaurus records available on Zenodo.

The records were filtered to include observations from 2024.

A total of 11,465 records was obtained for the selected period.

These records were grouped by date to calculate the number of reports recorded each day.

NASA DONKI Data

NASA's DONKI database was used to retrieve geomagnetic storm event records.

Unlike the GFZ API, the NASA request used an API key for authentication.

The 2024 data were downloaded month by month using Python and combined into a single dataset.

The resulting dataset contained 20 geomagnetic storm event records.

2.2 Python Data Processing

Python was used to download, organize, and process the scientific datasets.

The main libraries were:

- `requests` for API data retrieval.
- `pandas` for data processing.
- `pyarrow` for Parquet storage.
- `matplotlib` for graph generation.

The Kp measurements were checked for missing timestamps, duplicate values, and unexpected data ranges.

The original files were saved in JSON or CSV format, while the processed datasets were stored in CSV and Parquet formats.

For the correlation analysis, the maximum Kp value for each day was calculated.

The Aurorasaurus records were also grouped by day, and both datasets were matched using their UTC dates.

This resulted in 366 daily observations for the statistical comparison.

2.3 Correlation Analysis

Two correlation methods were used: Pearson and Spearman.

Pearson correlation measures the linear relationship between two variables.

Spearman correlation measures their rank-based relationship.

Both were calculated using the daily maximum Kp and the number of Aurorasaurus records reported on the corresponding date.

A positive coefficient indicates that higher values of one variable generally occur with higher values of the other variable.

However, correlation does not establish a direct causal relationship.

2.4 Power BI Analysis

The processed data were imported into Power BI to create interactive graphs and statistical summaries.

A date table was used to connect the daily observations with geomagnetic measurements and NASA storm-event records.

DAX calculations were used for summary statistics and daily comparisons.

The dashboard included monthly trends, a daily correlation scatterplot, geomagnetic activity throughout 2024, the May storm case study, and NASA storm-event comparisons.

The original October 2026 analysis was retained as a separate report page.

3 Results and Discussion

3.1 General Results for 2024

The Python analysis produced the results shown in Table 1.

Parameter	Result
Kp observations	2,928
Days analyzed	366
Average Kp	1.92
Maximum Kp	9.0
Storm-level intervals ($Kp \geq 5$)	97
Aurorasaurus observation records	11,465
Days with reports	318
Pearson correlation	0.4498
Spearman correlation	0.6181
NASA storm events	20

Table 1: Summary of the 2024 analysis.

Although the annual mean Kp was relatively low, the dataset contained several periods of elevated geomagnetic activity.

There were 97 three-hour intervals where Kp reached or exceeded 5.

This indicates that stronger geomagnetic disturbances occurred during certain periods of the year.

3.2 Monthly Activity and Aurora Reports

Figure 1 compares monthly average Kp values with the number of Aurorasaurus reports.

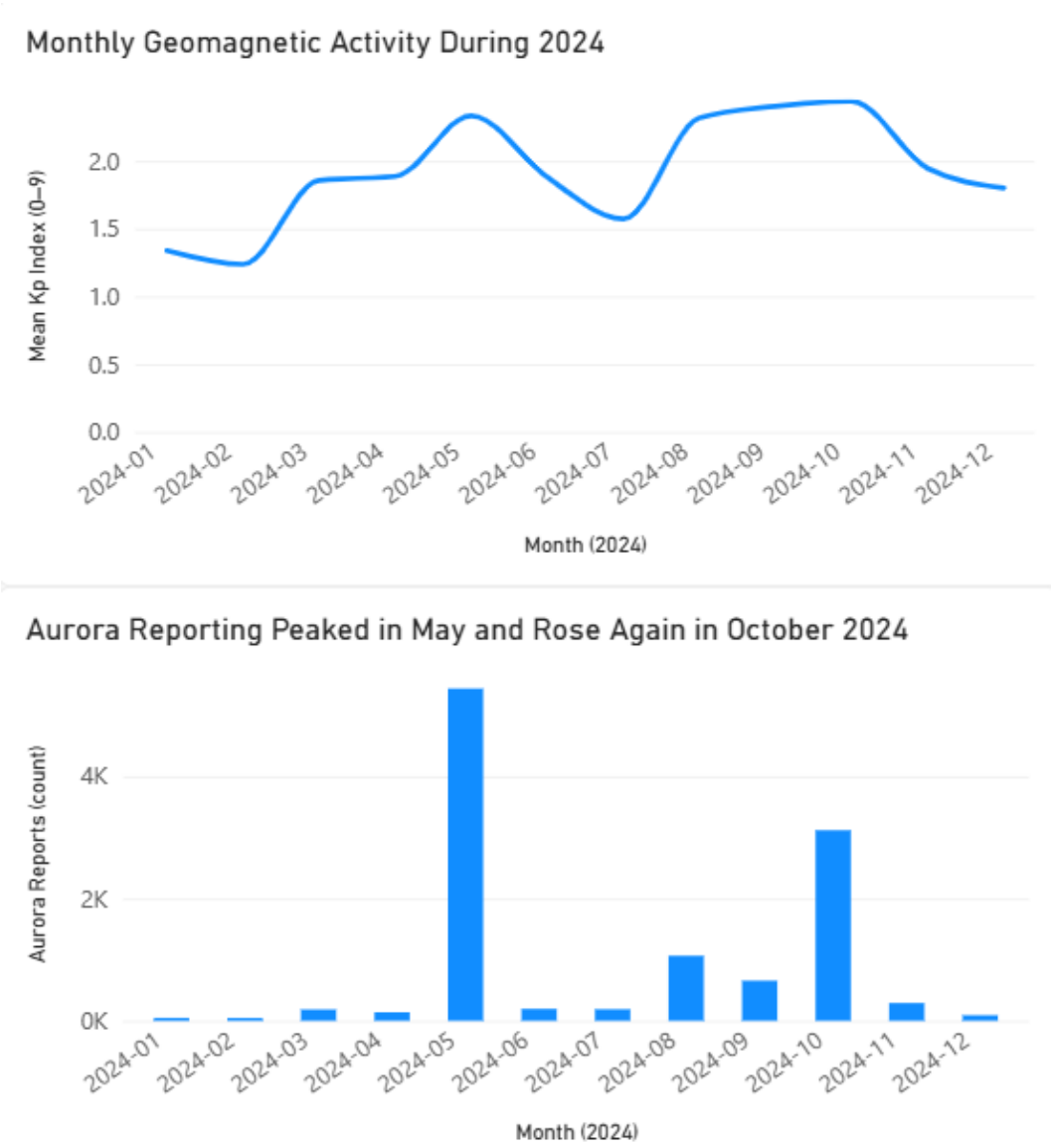


Figure 1: Monthly average Kp and aurora report counts during 2024. Sources: GFZ and Aurorasaurus.

The number of reported aurora observations was particularly high during May 2024.

A second noticeable increase occurred during October.

These periods were also associated with notable geomagnetic disturbances.

The monthly graph provides a general overview of the annual variation.

3.3 Correlation Between Kp and Aurora Reports

Figure 2 shows the daily maximum Kp compared with the number of aurora observation records.

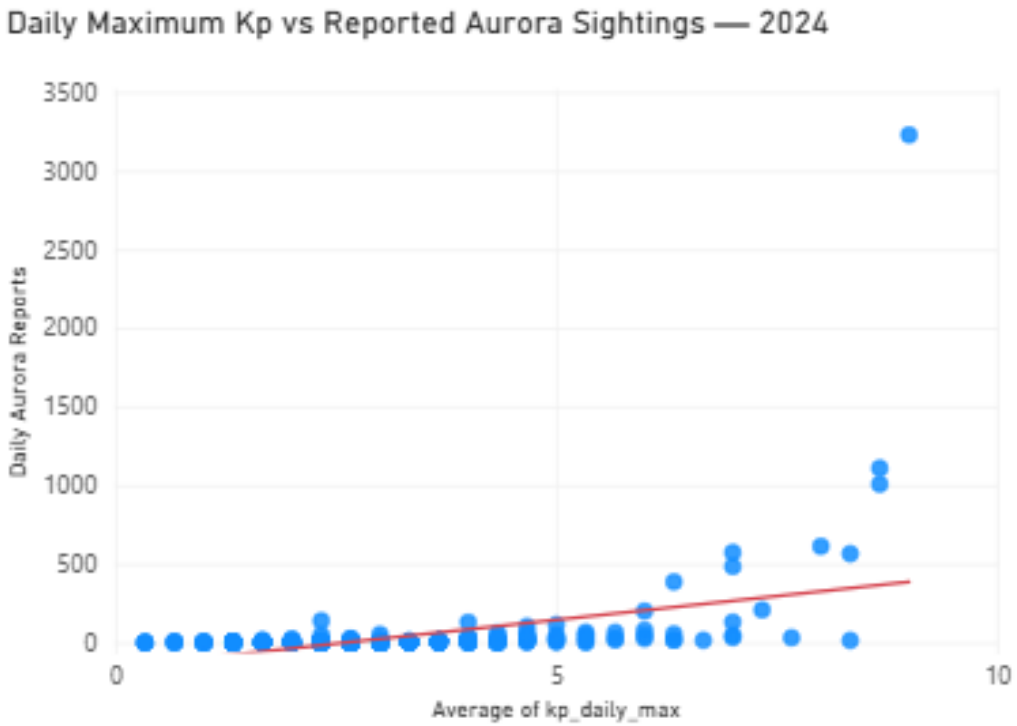


Figure 2: Daily maximum Kp versus Aurorasaurus report counts in 2024. Sources: GFZ and Aurorasaurus.

The Pearson correlation coefficient was calculated as:

$$r = 0.4498$$

The Spearman correlation coefficient was:

$$\rho = 0.6181$$

Both coefficients are positive.

The Pearson result suggests a moderate positive linear association, while the Spearman result indicates a stronger positive rank-based association.

The scatterplot also shows that some days with extremely high Kp values had much larger numbers of reports.

These extreme events influence the overall distribution.

The results suggest that stronger geomagnetic disturbances were generally associated with more recorded aurora observations.

3.4 Geomagnetic Activity During 2024

Figure 3 shows the daily maximum Kp throughout 2024.

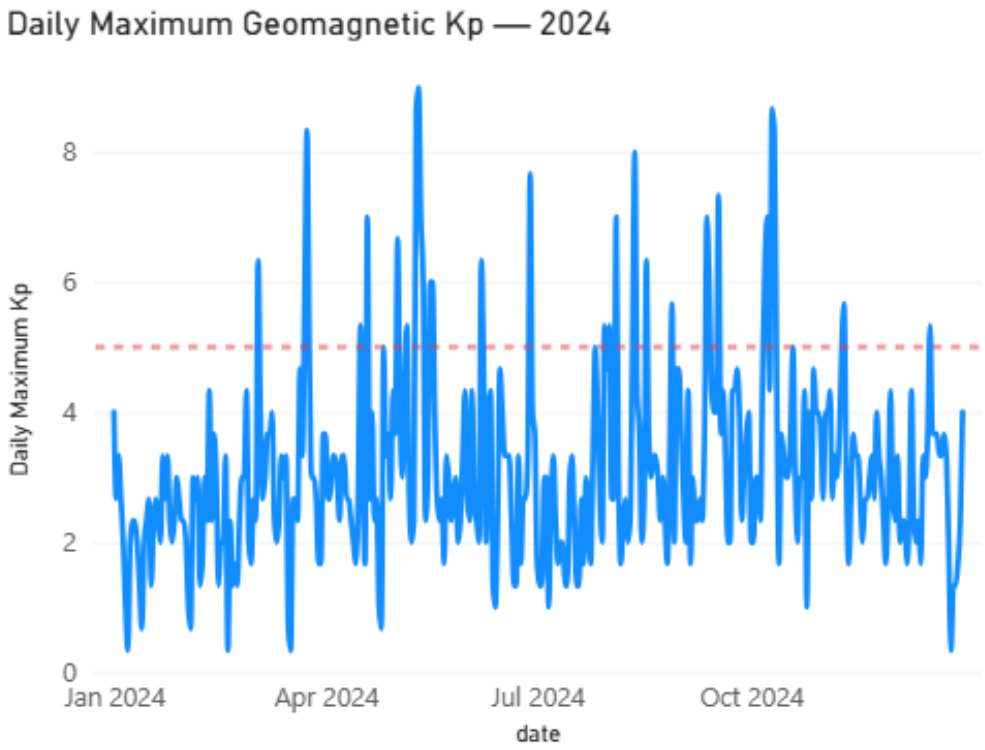


Figure 3: Daily maximum Kp during 2024. The dashed line indicates Kp = 5. Source: GFZ.

The graph shows relatively quiet conditions during many periods, interrupted by stronger geomagnetic disturbances.

Several peaks exceeded the storm-level threshold of Kp 5.

May and October were among the notable periods of increased activity.

3.5 May 2024 Geomagnetic Storm

The geomagnetic storm during May 10–11, 2024 was selected as a case study.

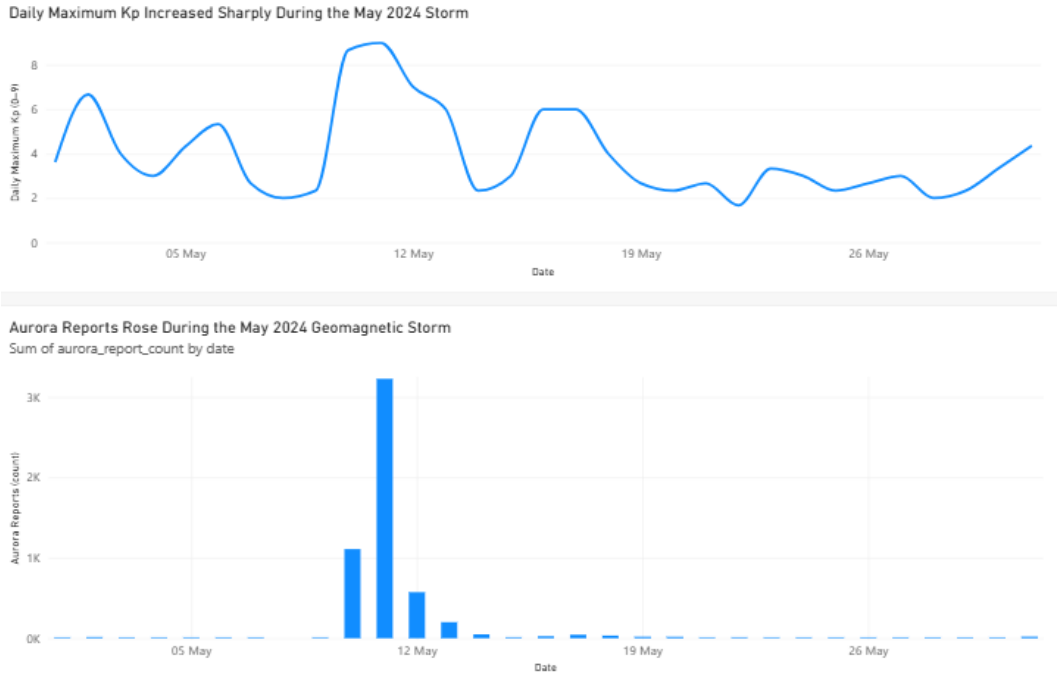


Figure 4: Daily Kp and reported aurora observations during May 2024. Sources: GFZ and Aurorasaurus.

During this period, geomagnetic activity increased significantly.

A large increase in aurora reporting occurred around the same time.

This event provides a clear example of how high geomagnetic activity can coincide with increased aurora reporting.

However, it does not establish that every individual report was caused by the same geomagnetic event.

3.6 NASA Geomagnetic Storm Analysis

NASA DONKI data were used as an additional source of geomagnetic storm information.

The 2024 dataset contained 20 storm events distributed across seven months.

The maximum Kp value reported within the retrieved event records was 9.0.

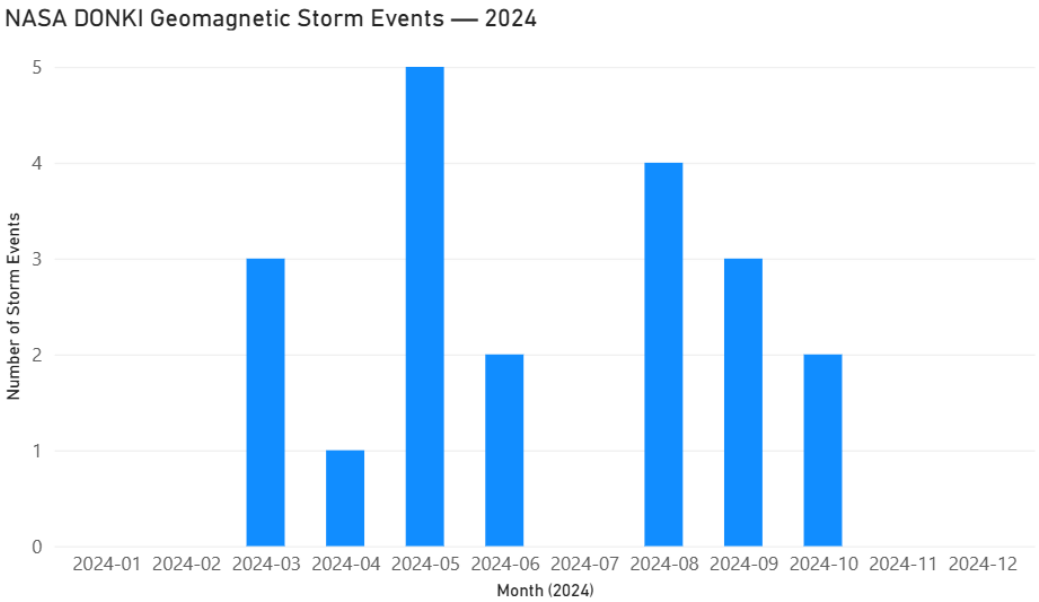


Figure 5: NASA DONKI geomagnetic storm events by month during 2024. Source: NASA DONKI.

The number of NASA storm events is different from the 97 storm-level Kp intervals measured by GFZ.

This is because NASA counts storm event records, while GFZ provides measurements every three hours.

3.7 Aurora Reports on NASA Storm Days

The dates of NASA geomagnetic storm events were compared with daily Aurorasaurus report counts.

Days on which NASA recorded the beginning of a storm were classified as storm-start days.

All other days were placed in a separate group.

The average number of reports per day was compared between these groups.

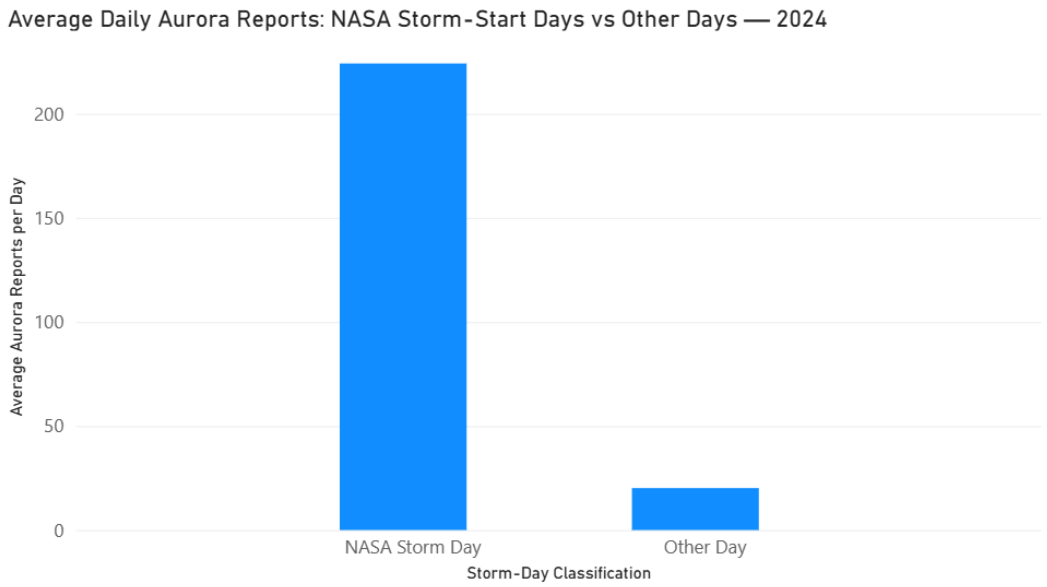


Figure 6: Average daily aurora reports on NASA storm-start dates compared with other dates during 2024. Sources: NASA DONKI and Aurorasaurus.

The Power BI results showed that the average daily report count was higher on NASA storm-start days than on other days.

This provides additional support for the positive association observed in the main analysis.

The total number of Aurorasaurus records calculated in Power BI was 11,465, matching the Python output.

4 Additional Case Study: October 2026

The project was initially developed using geomagnetic measurements from October 2026.

This dataset included 248 Kp observations covering 31 days.

The average Kp was 1.93, and the maximum Kp was 7.333.

Seven three-hour intervals reached $Kp \geq 5$.

The strongest disturbance occurred on October 10, 2026.

The same Python processing and Power BI visualization methods were used for this dataset.

This case study demonstrates that the workflow can be used for different observation periods.

The October 2026 data were not included in the 2024 aurora correlation calculation.

5 Limitations

The results should be interpreted with some limitations.

Aurorasaurus observations are citizen-submitted records. Their number depends on weather conditions, local darkness, geographic distribution, and observer participation.

The processing counted cleaned observation records, but did not independently confirm that every record was a positive aurora sighting.

Therefore, the report counts should not be interpreted as the total number of auroras that occurred

worldwide.

The correlation analysis also used daily aggregated data, which may hide variations within individual days.

Similarly, NASA storm-start dates identify when storm events began rather than every day a storm remained active.

Finally, statistical correlation does not prove causation.

Further analysis could include regional observations, weather conditions, or satellite-based measurements.

6 Conclusion

This project investigated the relationship between geomagnetic activity and reported aurora observations during 2024.

The analysis included 2,928 geomagnetic Kp measurements and 11,465 Aurorasaurus records.

Pearson and Spearman correlation coefficients of 0.4498 and 0.6181 indicated a positive statistical association between daily maximum Kp and the number of reported observations.

Overall, the results suggest that increased geomagnetic activity was generally associated with increased aurora reporting during 2024.

The main contribution of this project is the combination of scientific data retrieval, Python processing, statistical analysis, and Power BI visualization in a reproducible workflow.

References

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5. Project repository. https://github.com/mohamedayas16/Semester_2_CS_2026